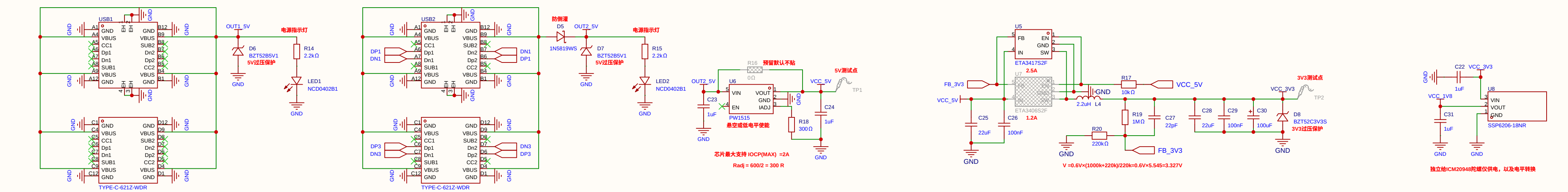
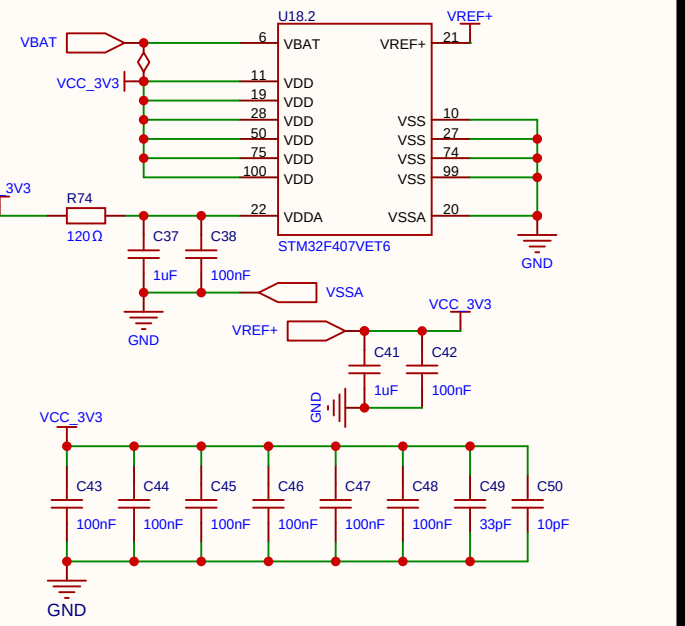


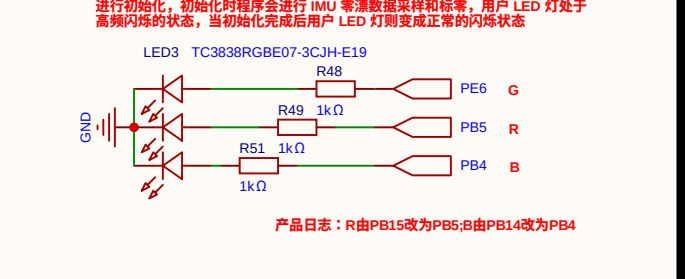
外部DC供电&反接 (DC 10~28V) DC-DC 10-28V to 5V 5AMAX (外部输出供电) DC-DC 10-28V to 5V 5AMAX (板载供电) DC-DC 10-28V to 5V 5AMAX (舵机供电)



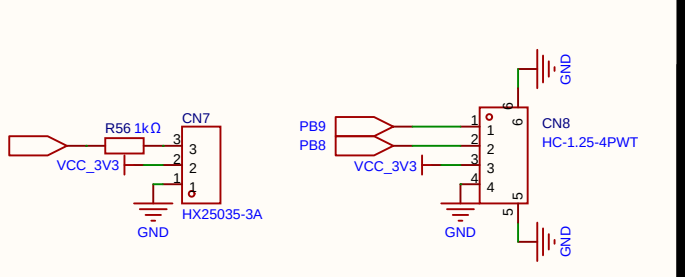
双层TYPE-C输出 5V 5AMAX 双层TYPE-C PC通信&烧录口 5V 2A MAX 过流短路保护 DC5V to 3V3 2.5AMAX 板载系统&模块供电



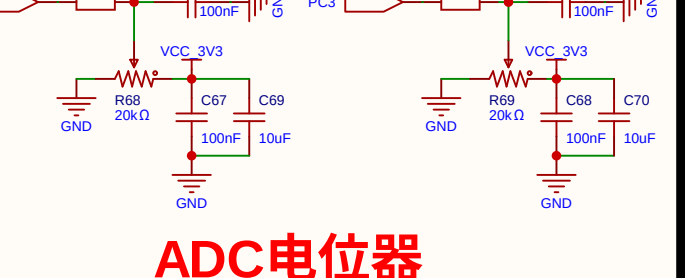
MCU外围电路&供电滤波



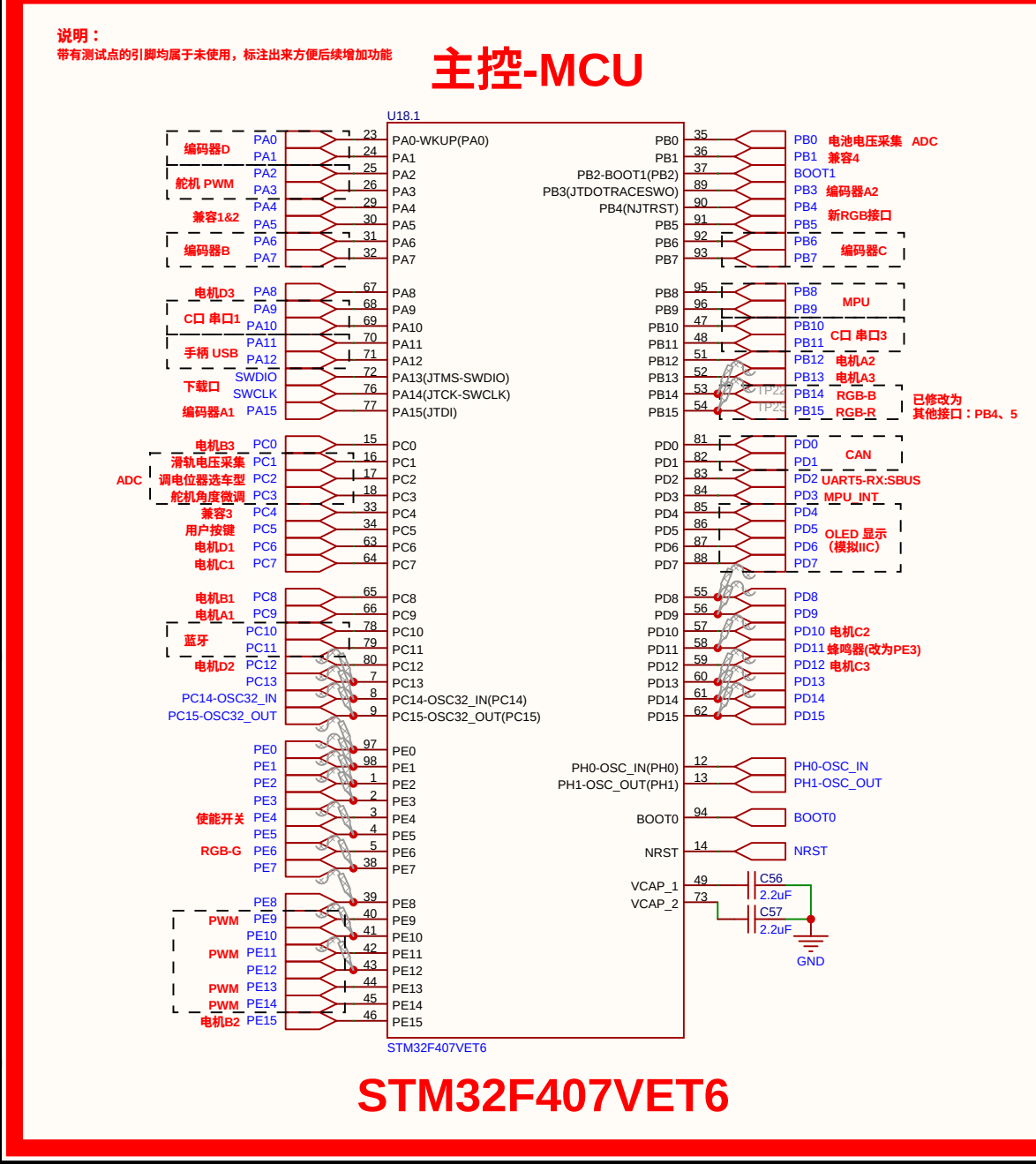
共阴极-RGB状态显示灯



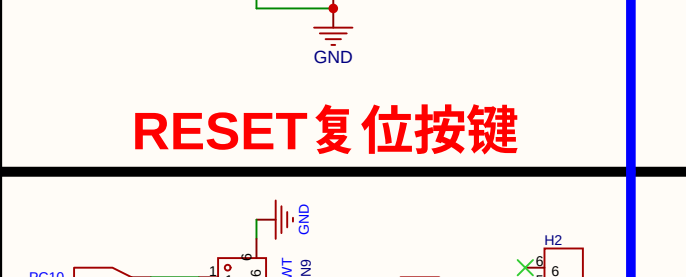
ADC接口预留&IIC接口



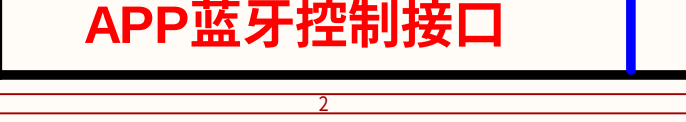
ADC电位器



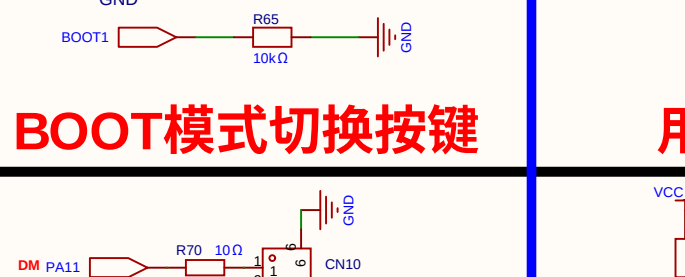
STM32F407VET6



RESET复位按键



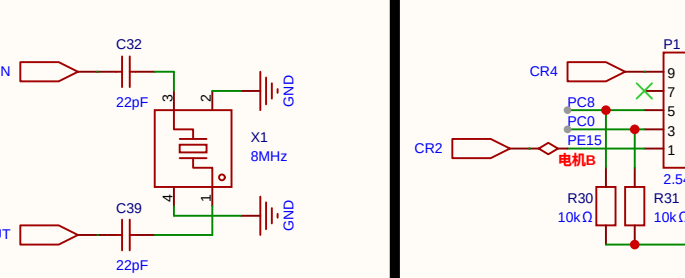
BOOT模式切换按键



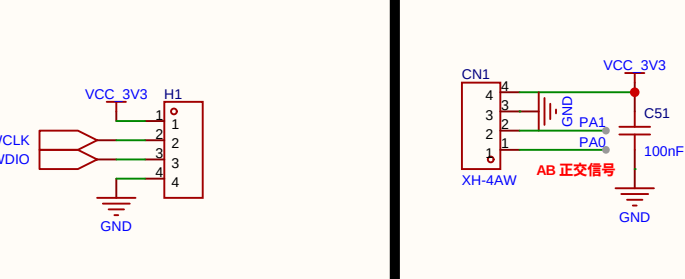
用户按键预留



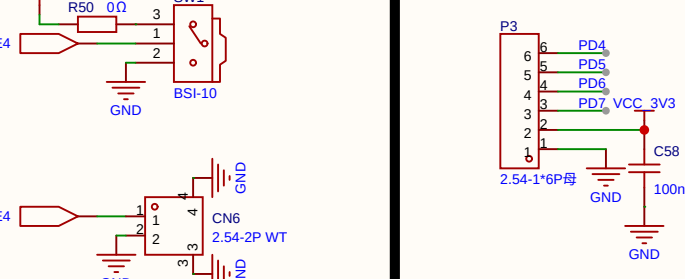
RUN指示灯



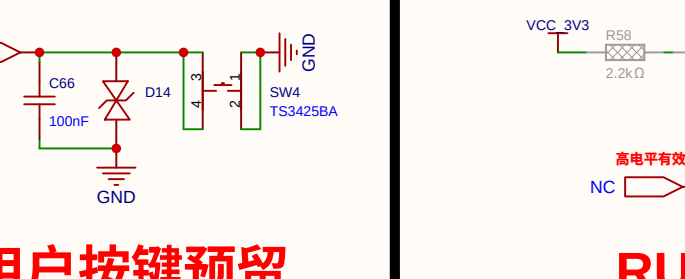
HSE 8MHZ时钟



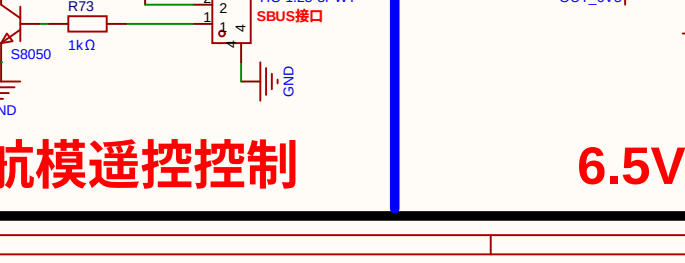
SWDIO烧录&调试



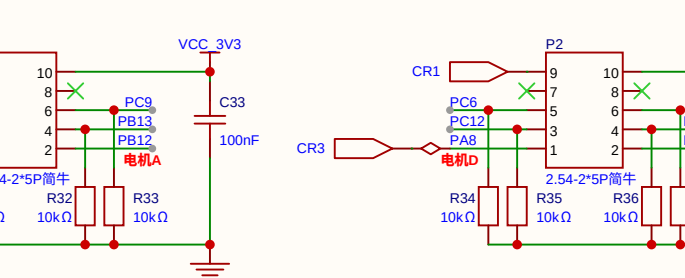
使能开关&急停接口



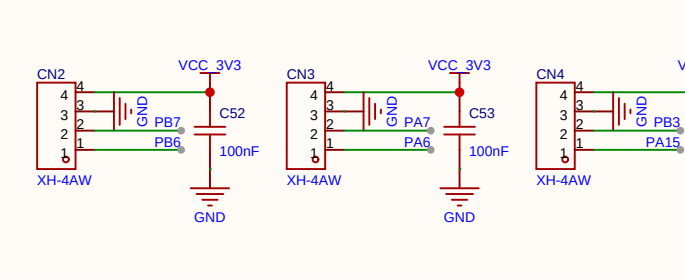
参数显示 OLED接口



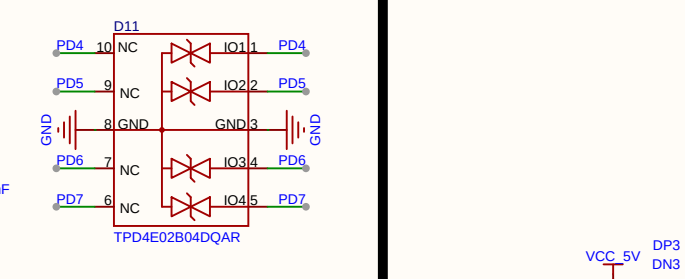
PC端通信电路(USART3)



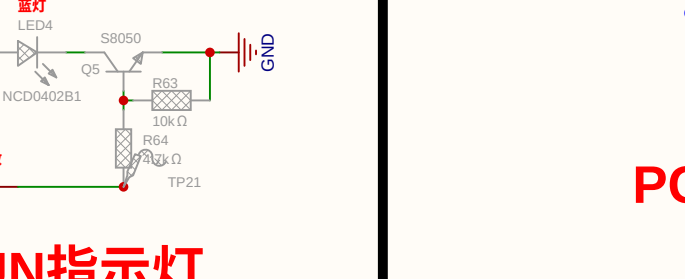
直流有刷电机模块驱动接口电路(D50A) * 4路 (4轮)



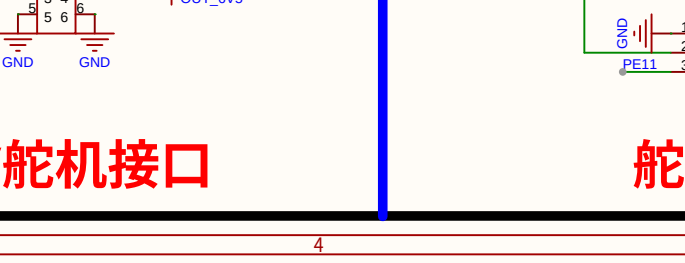
正交编码器采集 * 4路 (4轮)



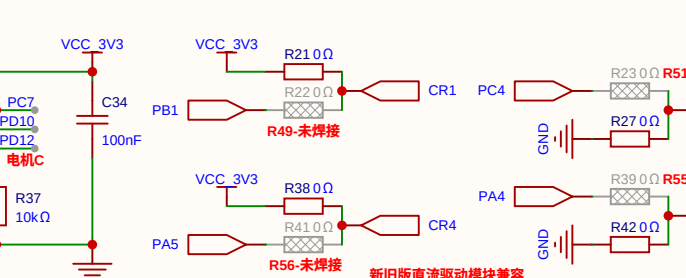
一键下载电路 (USART1)



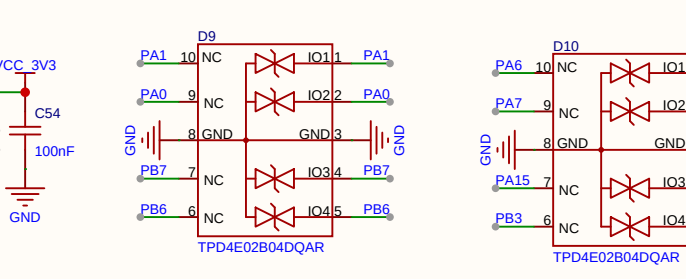
舵机&航模遥控 PWM * 4路



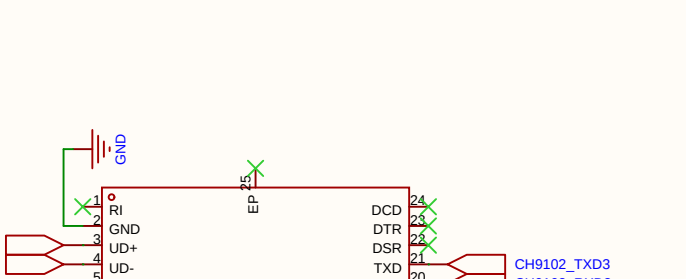
6.5V舵机接口



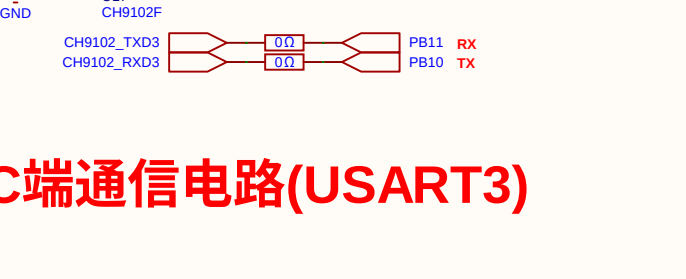
舵机&航模遥控 PWM * 4路



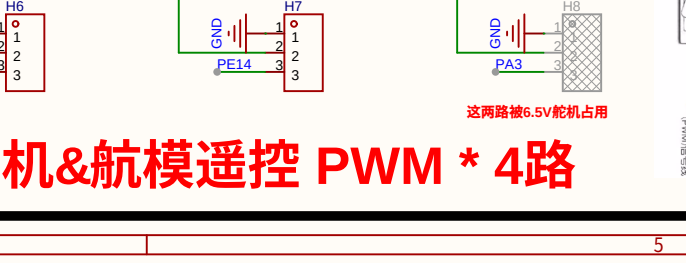
6.5V舵机接口



舵机&航模遥控 PWM * 4路



6.5V舵机接口



舵机&航模遥控 PWM * 4路



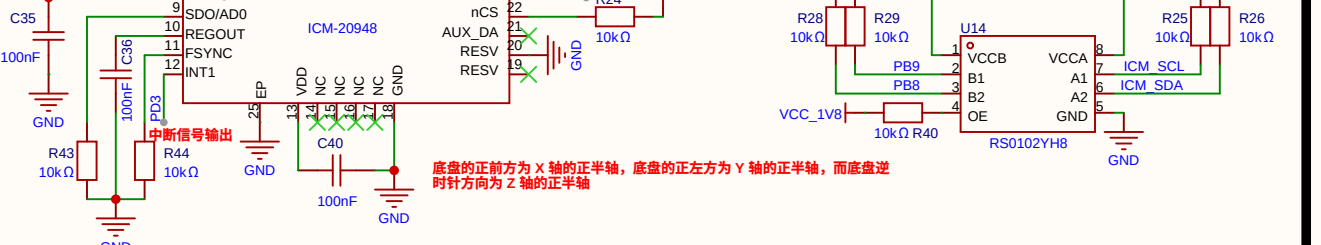
C50C_1V0版本说明

该驱动板适配ROS机器人/移动平台阿克曼差速、麦轮、四驱、全向轮等都适用。

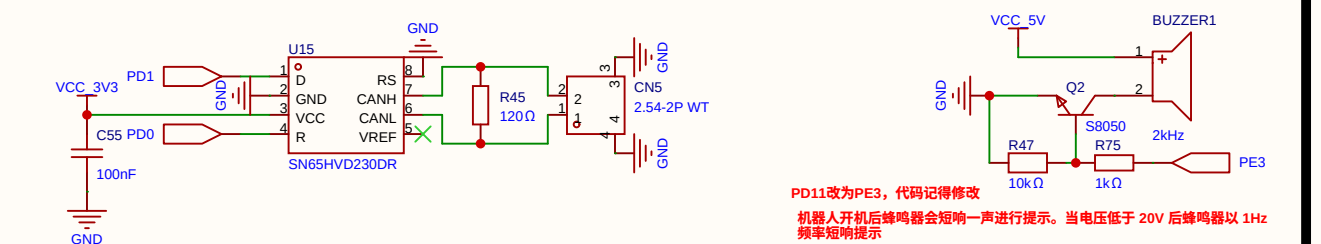
详细接口参数如右图所示：

名称	数量	C50C单片机主控板
主控芯片	1	STM32F407VET6
编码器接口	4	板载电机控制接口，用于实现电机闭环控制，星型计算
电机控制接口	2	可控制4个电机，通过阿克曼、差速、麦轮、轮式移动平台类型
PWM 3V3引脚	6	可用于控制舵机，或用于辅助定位
蓝牙模块接口	1	板载MPU6050，可用于辅助定位
串口通信接口	2	通过串口TTL与PC接口以USB直接传输，用于烧录、烧录程序
常用通信接口	3	CAN通信接口2个，I2C通信接口2个
蓝牙模块接口	2	用于APP远程遥控、在线测速
P2手抛接口	1	用于P2手抛
SWD接口	1	用于程序烧录、在线调试
电机接口	3	可用于调节产品型号，或作为ADC读取电压接口
SRV5接口	1	板载SRV5接口，可用于数据读取
人机交互相关接口	6	用户按键*1、复归按钮*1、蜂鸣器*1、LED*1、开关*1
按键接口	24	IO引脚*12、3V3*2、5V*2、GND*3
5V5A电源接口	2	用于给 Nano、Arduino、树莓派等开发板供电
电源输入接口	4	电压输入范围 10-28V
电机保护功能	1	过流保护、输入5A过流保护、USB接口5V保护、3.3V电压保护

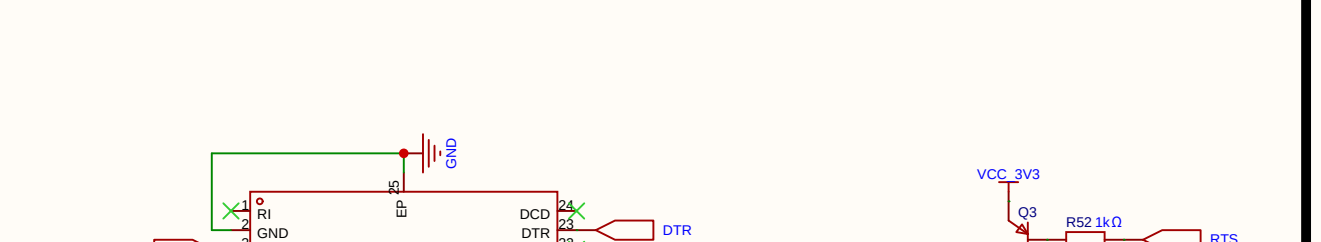
M3螺丝孔



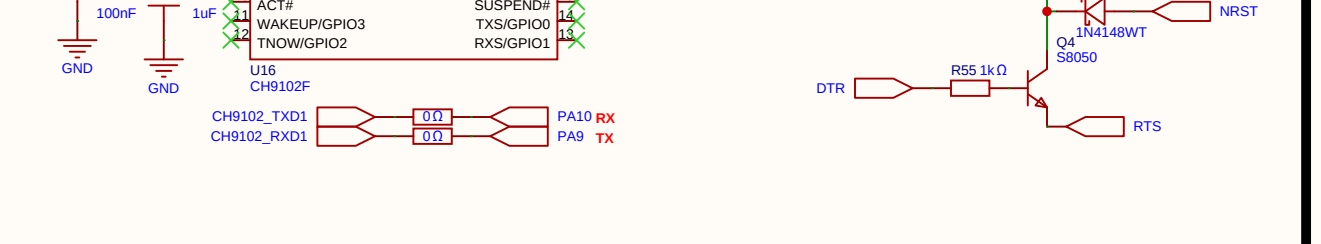
ICM20948陀螺仪&电平转换



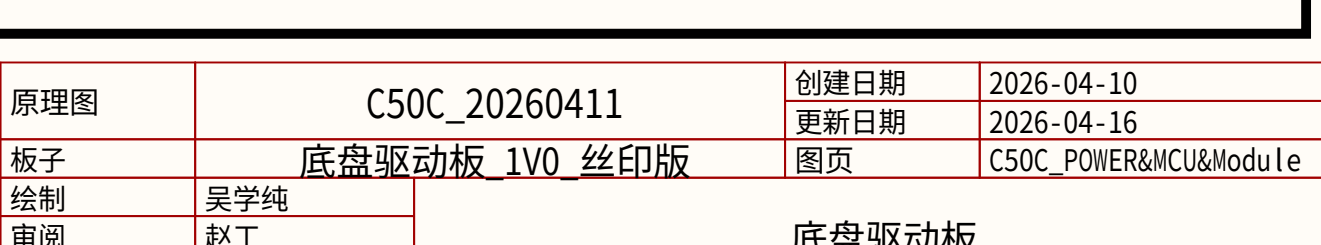
CAN总线 接口



蜂鸣器-指示响应



PC端通信电路(USART3)



一键下载电路 (USART1)



舵机&航模遥控 PWM * 4路

STM32F407VETx LQFP100

Categories: A-Z

Analogs

Timers

Connectivity

✓ CAN1

○ CAN2

○ ETH

○ FSMC

○ I2C1

○ I2C2

○ I2C3

○ SDIO

○ SPI1

○ SPI2

○ SPI3

UART4

○ USART1

○ USART2

○ USART3

○ USART6

○ USB_OTG_FS

○ USB_OTG_HS

Multimedia

Security

Computing

PA3 VSS PA4 VDD PA5 VDD PA6 VDD PA7 VDD PA8 VDD PA9 VDD PA10 VDD PA11 VDD PA12 VDD PA13 VDD PA14 VDD PA15 VDD PA16 VDD PA17 VDD PA18 VDD PA19 VDD PA20 VDD PA21 VDD PA22 VDD PA23 VDD PA24 VDD PA25 VDD PA26 VDD PA27 VDD PA28 VDD PA29 VDD PA30 VDD PA31 VDD PA32 VDD PA33 VDD PA34 VDD PA35 VDD PA36 VDD PA37 VDD PA38 VDD PA39 VDD PA40 VDD PA41 VDD PA42 VDD PA43 VDD PA44 VDD PA45 VDD PA46 VDD PA47 VDD PA48 VDD PA49 VDD PA50 VDD PA51 VDD PA52 VDD PA53 VDD PA54 VDD PA55 VDD PA56 VDD PA57 VDD PA58 VDD PA59 VDD PA60 VDD PA61 VDD PA62 VDD PA63 VDD PA64 VDD PA65 VDD PA66 VDD PA67 VDD PA68 VDD PA69 VDD PA70 VDD PA71 VDD PA72 VDD PA73 VDD PA74 VDD PA75 VDD PA76 VDD PA77 VDD PA78 VDD PA79 VDD PA80 VDD PA81 VDD PA82 VDD PA83 VDD PA84 VDD PA85 VDD PA86 VDD PA87 VDD PA88 VDD PA89 VDD PA90 VDD PA91 VDD PA92 VDD PA93 VDD PA94 VDD PA95 VDD PA96 VDD PA97 VDD PA98 VDD PA99 VDD PA100 VDD

原理图	C50C_20260411	创建日期	2026-04-10
板子	底座驱动板_1V0_丝印版	更新日期	2026-04-16
绘制	吴学纯	图页	C50C_POWER&MCU&Module
审核	赵工		
	吴学纯	底座驱动板	
	赵工	版本	页 1 共 1
		尺寸	1 1
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